

EECE562 Project: Unsupervised Brain MRI Tumor Segmentation by K-means and GMM

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Abstract—In many clinical applications, brain MRI segmentation is a basic task, which provides an essential reference for clinicians in the diagnosis. But due to the high labor cost, the analysis of brain MRI data requires the computerized method. In this project, we use two segmentation methods K-means and GMM on the public Brain Tumor Segmentation Challenge Datasets (BraTS) 2018. we randomly selected 10 samples and evaluate them, showing that GMM achieved better results on unsupervised segmentation, but both methods failed to segment smaller tumors.

Index Terms—MRI, segmentation, unsupervised learning, K-means, GMM, EM algorithm.

I. INTRODUCTION

In recent years, advances in brain magnetic resonance imaging (MRI) have provided a large amount of data with high quality, which has also allowed people to explore brain anatomy or brain lesions in a more refined and accurate manner. However, analysis of these large and complex MRI datasets is complex and time-consuming for clinicians, because they must manually extract important information. Thus, the analysis of brain MRI data requires the computerized method to improve disease diagnosis.

In many clinical applications, brain MRI segmentation is a basic task because it extracts the patient-specific important clinical information and diagnostic features [1], which provides an essential reference for clinicians in the diagnosis of diseases and follow-up treatment and surgical planning. In this project, we first introduce 3D brain MRI preprocessing steps including normalization and convolution. Then, we evaluated two methods, K-means and GMM, for brain MRI segmentation. Finally, we highlight the differences between them and discuss their advantages and limitations.

II. METHODS

A. Materials

We used the public Brain Tumor Segmentation Challenge Datasets (BraTS) 2018 [2]. It had 285 patients with LGGs and HGGs from 19 institutions where different protocols were used to acquire the MR data. HGGs had 210 samples and LGGs had 75 samples. Each sample had T1 (T1), T1W + contrast (T1c), T2-weighted (T2) and FLAIR images. The size of each image is $240 \times 240 \times 155$. The brain tumors were segmented manually by neuroradiologists. Annotations include 3 tumor subregions—the enhancing tumor (ET), the peritumoral edema (ED), and the necrotic and non-enhancing

tumor core (Net). The annotations were combined into 3 nested subregions—whole tumor (WT), tumor core (TC), and enhancing tumor (ET).

B. MRI preprocessing

Before applying the segmentation method, we need to normalize MR images for posterior analysis. The contrast in the images of the 4 modalities was different, so we used the z-score method to normalize the images of each modality. Each image was normalized to zero mean and unit standard deviation.

C. Feature extraction

Feature extraction comprises the process of obtaining new features from the MRI images to improve the discriminative power between different tissues in the brain. The information of each voxel in a medical image is determined not only by its intensity but also by the intensities of its surrounding voxels. Liu proposed that in 2D brain MRI segmentation, using the average value of the 3×3 neighborhood of each image pixel for segmentation could effectively reduce the impact of noise on the accuracy of clustering segmentation [3]. Our project focused on the 3D brain MRI images, so we used a local 3D neighborhood of $3 \times 3 \times 3$ voxels. The feature of each voxel was the average value of its 3D neighborhood.

Thus, a set X of 4 features was obtained for each patient, consisting on the following images:

$$X = (\mu T1, \mu T1c, \mu T2, \mu Flair)$$

where μ , referred to the mean feature of the corresponding images.

D. Unsupervised voxel classification

MRI segmentation aims to divide an image into a set of homogeneous regions of similar grayscale. The segmentation result is a labeled image that each label identifies each homogeneous region. So the essence of brain MRI segmentation is to cluster the different homogeneous regions in the brain.

The BRATS 2018 dataset comprises 4 classes to be segmented: class 0 represents background, class 1 represents Net, class 2 represents ED, and class 4 represents ET. WT includes Net, ED, and ET. In addition, we assumed that 4 tissues existed in a brain, which was white matter (WM), gray matter (GM), cerebro-spinal fluid (CSF), and tumor. Since

WM and GM looked similar in images, we grouped them together as WM&GM. Furthermore, due to the area inside the brain being bright on the image, whereas the outer area was dark, the voxels at the edge of the brain had special features after convolution, we named this class Edge. Therefore, in this project, we classified voxels into 5 classes: WT, WM&GM, CSF, Edge, and Background.

Let $X = \{x_1, x_2, \dots, x_N\}$ be the set of voxels to be classified, where $x_n \in \mathbb{R}^4$ represents a feature vector of 4 dimensions for voxel n , and $N = 240 \times 240 \times 155$. Let $J = \{J_1, J_2, \dots, J_M\}$ be the set of all possible classes for the segmentation where $M = 5$. Let $Y = \{y_1, y_2, \dots, y_N\}$ a segmentation of the brain volume, where $y_n \in J$.

E. K-means

K-means is an unsupervised clustering based on a distance minimization criterion. Its aim is to divide the data space X into M clusters, $J = J_1, J_2, \dots, J_M$ so that each observation of X belongs to the cluster with the nearest centroid [4]. The distance criterion minimized by K-means is

$$\operatorname{argmin}_j \sum_{j=1}^M \sum_{x_n \in J_j} \|x_n - \mu_j\|^2 \quad (1)$$

Where μ_j is the center of cluster j and for our project $M = 5$.

F. GMM-EM

Gaussian mixture model (GMM) clustering is a model-based classification algorithm whose aim is to find the maximum likelihood parameters of a Mixture of Gaussian distributions that best fit the data to be classified. In this method, we first regard each homogeneous region as a Gaussian distribution, which is a Gaussian mixture model [5]. Then, use the expectation-maximization (EM) algorithm to predict the parameter of each component, and cluster each voxel.

1) *GMM*: A Gaussian mixture model (GMM) is a weighted sum of Gaussian distributions of M components. It is a probabilistic model assuming that the voxel data points are generated from a mixture of Gaussian components. The probability distribution function of GMM can be written as:

$$h(x_n) = \sum_{j=1}^M \pi_j h_j(x_n | \mu_j, \Omega_j) \quad (2)$$

where each data point $x_i \in \mathbb{R}^d$ comes from candidate distribution:

$$h_j(x | \mu_j, \Omega_j) = (1/2\pi)^{d/2} |\det(\Omega_j)|^{-1/2} e^{-1/2(x - \mu_j)^T \Omega_j^{-1} (x - \mu_j)} \quad (3)$$

with probability π_j , and

$$\sum_{j=1}^M \pi_j = 1 \quad (4)$$

The parameters of a GMM are: the mean vectors of each component μ , the covariances matrices of each component Ω , and the weights associated with each component π .

2) *EM*: In the EM algorithm, first, make an initial guess about which voxel belongs to which component via the initial means and variances of the components, then iteratively refine this guess and make this guess converge to an optimal solution.

In this project, we use the result of K-means as the initial mean of each component, use unit variance as the initial variance of each component and use $1/M$ as the initial weight associated with each component. And estimate parameter $\theta = (\pi_{1:M}, \mu_{1:M}, \Omega_{1:M})$ by given voxel data $x_{1:N}$ at each iteration.

After initialization, define the missing data:

$$Z_{ij} = I\{x_i \text{ comes from model } j\} \quad (5)$$

where $\sum_{j=1}^M Z_{ij} = 1$.

In the Expectation step (E-step), we need to estimate the probability that a given observation x_i belongs to a given component Z_{ij} :

$$Pr(Z_{ij} = 1 | x_i, \theta_j) = \frac{\pi_j P(x_i | \theta_j)}{\sum_{j=1}^M \pi_j P(x_i | \theta_j)} = W_{ij} \quad (6)$$

and calculate the conditional expectation Q :

$$Q(\theta | \theta^t) = E[\log f(X, Z | \theta) | X = x_{1:N}, \theta^t] \quad (7)$$

where θ^t is the old parameter value, and θ is the new parameter value. Then, compute the complete log-likelihood:

$$\log f(X, Z | \theta) = \sum_{i=1}^N \sum_{j=1}^M Z_{ij} (\log \pi_j + \log h_j(x_i | \mu_j, \Omega_j)) \quad (8)$$

Plug in the two equations (7) (8), and we can derive the conditional expectation Q :

$$Q(\theta | \theta^t) = \sum_{i=1}^N \sum_{j=1}^M W_{ij}^t \log \pi_j + \sum_{i=1}^N \sum_{j=1}^M W_{ij}^t (-0.5 \log \det(\Omega_j) - 0.5(x_i - \mu_j)^T \Omega_j^{-1} (x_i - \mu_j)) \quad (9)$$

In the Maximum step (M-step), we maximize the value of Q to find the optimal parameter value:

$$\theta^{t+1} = \operatorname{argmax}_{\theta} Q(\theta | \theta^t) \quad (10)$$

$$\text{set } \frac{\partial Q(\theta | \theta^t)}{\partial \pi^t} = \frac{\partial Q(\theta | \theta^t)}{\partial \mu^t} = \frac{\partial Q(\theta | \theta^t)}{\partial \Omega^t} = 0 \text{ to get } \theta^{t+1}$$

where $\theta^{t+1} = (\pi^{t+1}, \mu^{t+1}, \Omega^{t+1})$:

$$\pi_j^{t+1} = \frac{\sum_{i=1}^N W_{ij}^t}{N} \quad (11)$$

$$\mu_j^{t+1} = \frac{\sum_{i=1}^N W_{ij}^t x_i}{\sum_{i=1}^N W_{ij}^t} \quad (12)$$

$$\Omega_j^{t+1} = \frac{\sum_{i=1}^N W_{ij}^t (x_i - \mu_j^{t+1})(x_i - \mu_j^{t+1})^T}{\sum_{i=1}^N W_{ij}^t} \quad (13)$$

Because the clustering results of GMM were affected by the initial value, we used the centers μ_j' s obtained by K-means as the initial mean value μ_0 of the EM algorithm to promote it to converge to the global maximum.

G. Remove WM&GM, Edge and Background classes

After clustering, as shown in Fig. 1 the voxels were classified into 5 classes (WT, WM&GM, CSF, Edge, Background), but we were only interested in the WT class, so the voxels of other 4 classes should all be assigned to the Background class.

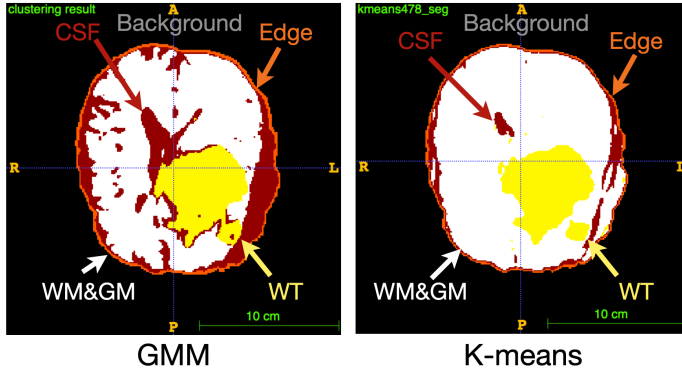


Fig. 1. Clustering result

First, it was necessary to determine the label of the Background class. For a 3D image, 5 slices were randomly selected, and the voxels located in the 4 corners of the 5 slices were most likely to belong to the Background class, so the mode of the labels of these 20 voxels was regarded as the label of the Background class.

Next, remove the WM&GM and Edge classes. In a brain, white matter and gray matter accounted for the largest volume and the edge occupied the smallest volume. Therefore, in addition to the voxels of the Background class, the class with the largest number of voxels was assigned as WM&GM class, and the class with the smallest number of voxels represented the Edge class. We let the labels of the voxels of these two classes be the labels of the Background.

However, the last two classes, CSF and WT, were difficult to distinguish depending on the number of voxels or the probability of the class. In our experiments, sometimes there were more voxels in the CSF class, but sometimes more voxels were labeled by the WT class. Although the number of voxels contained in these two classes was similar, the images represented by their voxels were significantly different. In this step, we used human eyes to judge the CSF and WT classes. Specifically, as shown in Fig. 2 and Fig. 3, each image only contained the voxels of one class, and distinguish tumor class from CSF class by the shape of the voxels. It was evident that image 1 represented the WT class and image 2 was the CSF class. Thus, image 1 was our final segmentation result.

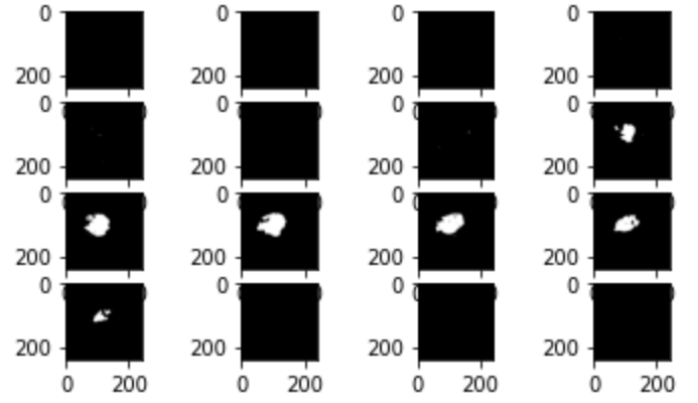


Fig. 2. Image 1

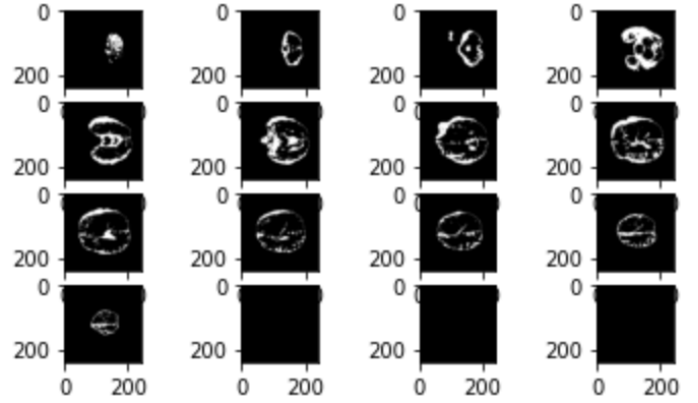


Fig. 3. Image 2

H. Experiment design

In BraTS 2018, we randomly selected 10 samples from HGGs and segment whole tumor by K-means and GMM. Each sample was trained and segmented independently, and the ground truth annotations were only used in calculating the evaluation metrics.

We evaluated the two methods with the commonly accepted evaluation metrics for brain tumor segmentation - including Dice score (Dice), sensitivity (Sens.) and precision (Prec.). These terms are defined as follows:

$$Dice = \frac{2 \times |GT \cap PM|}{|GT + PM|} \quad (14)$$

$$Sens. = \frac{|TP|}{|TP + FN|} \quad (15)$$

$$Prec. = \frac{|TP|}{|TP + FP|} \quad (16)$$

The ground truth annotation of brain tumors provided in the BraTS datasets is GT; PM is the predicted mask of the tumor by our methods; TPs are the true positive tumor voxels; FPs the false positive tumor voxels; FNs are the false negative background voxels and TNs are the true negative background voxels.

III. RESULT AND DISCUSSION

Table I and II show the results of brain MRI tumor segmentation obtained from random 10 patients, measured by the Dice score, sensitivity, and precision respectively.

TABLE I
TUMOR SEGMENTATION BY K-MEANS

patient	Dice	Sens.	Prec.
1	0.8855	0.8773	0.8939
2	0.8872	0.9333	0.8454
3	0.4518	0.8958	0.3020
4	0.8864	0.8955	0.8775
5	0.9008	0.8403	0.9708
6	0.9044	0.9028	0.9061
7	0.0037	1.0000	0.0019
8	0.0031	0.0029	0.0035
9	0.8709	0.7823	0.9821
10	0.3876	0.9429	0.2439

TABLE II
TUMOR SEGMENTATION BY GMM-EM

patient	Dice	Sens.	Prec.
1	0.8890	0.8648	0.9145
2	0.8921	0.9437	0.8459
3	0.4469	0.8685	0.3009
4	0.8406	0.9294	0.7673
5	0.9148	0.8476	0.9936
6	0.8907	0.9461	0.8146
7	0.0086	1.0000	0.0043
8	0.3479	0.6611	0.2361
9	0.9286	0.9177	0.9397
10	0.4845	0.9770	0.3221

From Table I, K-means method had the best in Dice at 90.44% with Sens. of 90.28%. From Table II, GMM-EM had the best in Dice at 92.86% with Sens. of 91.70%. For patients 1, 2, 4, 5, 6, and 9, the Dice scores for K-means and GMM were around 90%, which means for 60% of the samples, both K-means and GMM could accurately segment the tumor. For a specific sample, patient 5, K-means had the Dice of 90.08% and Sens. of 84.03%; GMM had the Dice of 91.48% and Sens. of 84.76%. As shown in Fig. 4 and 5, the predicted masks of K-means and GMM were similar to the ground truth (Fig. 6). Both metrics and images indicated that both two methods could effectively segment the whole tumor for patient 5.



Fig. 4. Predicted mask of K-means for patient 5

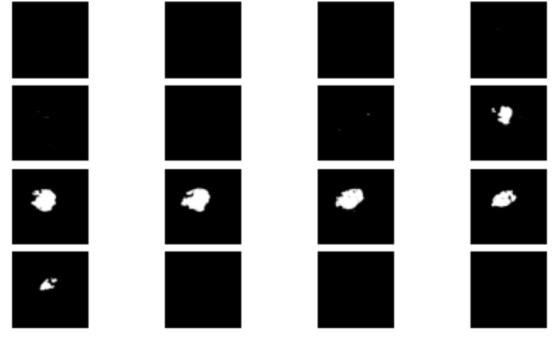


Fig. 5. Predicted mask of GMM for patient 5

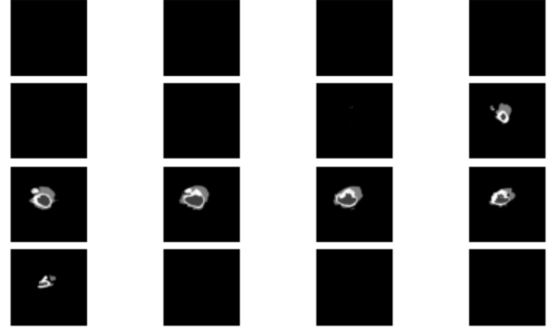


Fig. 6. Ground Truth

For most samples, the Dice scores of GMM were slightly higher than that of K-means. In particular, for patient 8, K-means had a Dice score of 0.31%, and its result was shown in Fig. 7, implying that K-means failed to segment the whole tumor. However, the GMM's Dice score was 34.79%, which was much higher than that of K-means. Its result was shown in Fig. 8, which indicates the GMM could segment parts of the tumor.

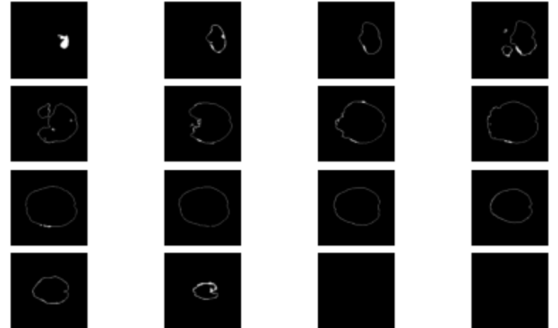


Fig. 7. Predicted mask of K-means for patient 8

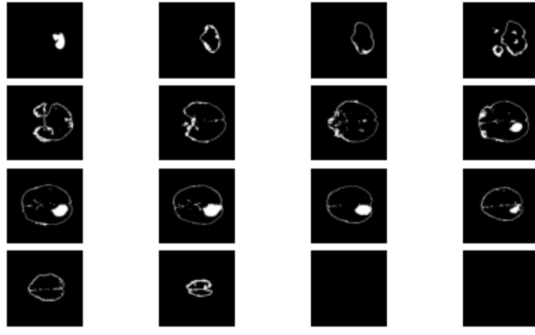


Fig. 8. Predicted mask of GMM for patient 8

In order to prevent this low score of K-means from the wrong operation of removing WM&GM, CSF and Background classes, we also showed the results of K-means clustering in Fig. 9. It clearly showed that K-means failed to segment the tumor, but GMM successfully outlined the shape of the tumor, and GT represented the ground truth segmentation of patient 8.

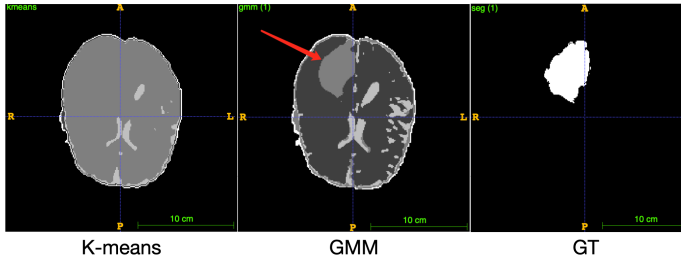


Fig. 9. Clustering result and GT for patient 8

According to Table I and II, both K-means and GMM had low scores for patient 7. As shown in Fig. 10, the tumor was too small. Thus, neither approach was able to separate the tumor from the brain when the tumor was too small.

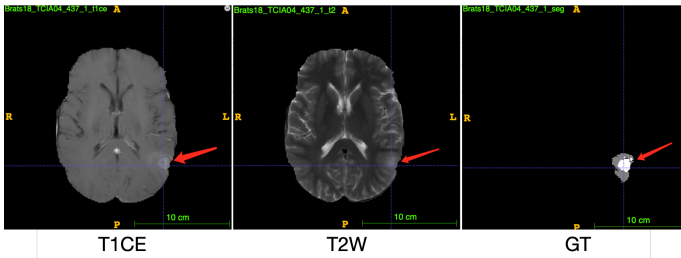


Fig. 10. T1CE, T2W and GT for patient 7

The two main problems of the EM algorithm were slow convergence and possible convergence to local optimum. We use GPU acceleration by PyTorch to greatly reduce our run time. The Dice scores of GMM were mostly higher than those of K-means, which means that using the result of K-means as the initial mean value of the EM algorithm could effectively prevent the EM algorithm converge to the unwanted local optimum.

IV. LIMITATIONS AND FUTURE WORK

The failure to segment smaller tumors was actually expected because the number of clusters was not large enough to segment smaller tissues. A larger number of clusters could solve this problem, but it might bring a new problem, which was difficult to decide which clusters should be removed after classification. Moreover, our method does not perform the segmentation independently by the computerized program. In the step of removing CSF, we need to distinguish CSF and tumor manually.

In the future, we will study how to use dimensionality reduction to train multiple samples at the same time, and how to use GAN to do unsupervised segmentation.

V. CONCLUSION

We use K-means and GMM for unsupervised brain tumor segmentation. Our results on public brain tumor datasets show that GMM achieved better results on unsupervised segmentation, but both methods failed to segment smaller tumors.

VI. APPENDIX

Our code and dataset can be found in the google drive.
https://drive.google.com/drive/folders/16aFjy3HCQm1_i2KrjFPP5P9QAmeBTiHu?usp=sharing

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